Report 6: Fine-Tuning Analysis — Will Domain Adaptation on AVSpeech Improve VSP-LLM?

Document Classification: Technical Research Report **Date:** February 18, 2026 **Scope:** Whether and how fine-tuning/QLoRA on diverse YouTube-like data (AVSpeech) would improve the model's real-world performance **Current Performance Baseline:** Mean WER 67.0%, Corpus WER 125.5% on english_1k (860 segments)

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1. Executive Summary

The VSP-LLM model's 67% WER on real-world YouTube content (vs. 25.4% on the LRS3 benchmark) is primarily caused by **domain mismatch** — the model was trained exclusively on curated TED talks. Fine-tuning on diverse, YouTube-sourced data like AVSpeech would address this directly.

Expected improvement: 15-25 WER points (from ~67% to ~42-52%), with the strongest gains coming from unfreezing the AV-HuBERT visual encoder, which learns to handle diverse head poses, lighting conditions, and speaker appearances. However, fine-tuning is **not a complete solution** — hallucination (an architectural issue) and short-segment failure (a context issue) persist regardless of training data.

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2. Architecture: What Gets Trained

The VSP-LLM model consists of four trainable components. Understanding what each does is critical for predicting fine-tuning impact.

2.1 Component Breakdown

Component	Parameters	Trained by Default?	Purpose
AV-HuBERT encoder	~315M	Frozen first 18k steps, then unfrozen	Extracts visual features from lip movement video
avfeat_to_llm (linear layer)	1024 x 4096 = ~4.2M	Always trainable	Maps visual features to LLaMA embedding space
LoRA adapters (Q,K,V projections)	rank 16 x 32 layers x 3 = ~4.2M	Always trainable	Adapts LLaMA's attention for lip-reading

Component	Parameters	Trained by Default?	Purpose
LLaMA-2-7B base	~6.7B	**Never trained** (4-bit quantized)	Language generation backbone

Total trainable: ~323M when encoder unfrozen, ~8.4M when frozen.

2.2 Current Training Data

All components were trained on **LRS3** (Afouras et al., 2018): - **433 hours** of TED and TEDx talks - **Curated conditions:** frontal-facing speakers, studio lighting, professional cameras - **Standard vocabulary:** academic English, well-articulated - **Clean audio-derived transcriptions:** high-quality labels

2.3 The Domain Gap

The english_1k test data (YouTube) differs from LRS3 (TED) in every dimension:

Dimension	LRS3 (Training)	english_1k (Test)	Impact on WER
Head pose	Frontal, static	Variable, moving	High — encoder never learned non-frontal features
Lighting	Professional studio	Variable, often poor	High — encoder features degrade
Speaker diversity	TED presenters	Global YouTube creators	Medium — lip shapes vary
Vocabulary	Academic English	Casual, slang, technical, brands	Medium — LLM generates wrong words
Speech style	Rehearsed, clear	Natural, fast, mumbled	Medium — temporal patterns differ
Video quality	Professional cameras	Webcams, phones, screen recordings	Medium — encoder resolution sensitivity

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3. Expected Impact by Component

3.1 LoRA Adapters Alone (~4.2M trainable params)

These small adapter matrices shift LLaMA's attention patterns for lip-reading. Training them on diverse data would: - **Help:** Broader vocabulary prior (casual speech, brand names, slang) - **Help:** Better handling of sentence fragments and informal structure - **Not help:** Visual feature

quality (the encoder is frozen) - **Not help:** Hallucination (fundamentally an LLM architecture issue)

The paper's "freeze" variant (LoRA only, encoder frozen) achieves 26.7% WER on LRS3.

Expected improvement on out-of-domain data: 3-8 WER points (from ~67% to ~59-64%)

Rank-16 LoRA has very limited capacity — only ~4.2M parameters adapting a 6.7B model. It can nudge vocabulary priors but cannot fundamentally change language generation behavior.

3.2 LoRA + Bridge Layer (~8.4M trainable params)

The bridge layer (`avfeat_to_llm` , 1024 -> 4096 linear) translates visual features into LLaMA's embedding space. Training it on diverse inputs helps it handle the different feature distributions that non-ideal visual conditions produce.

Expected improvement: 5-10 WER points (from ~67% to ~57-62%)

3.3 LoRA + Bridge + Encoder Unfrozen (~323M trainable params) — RECOMMENDED

This is the "FT" variant from the paper. The AV-HuBERT encoder learns to extract meaningful features from diverse visual conditions — the primary bottleneck.

Evidence from the paper: - Freeze (LoRA only): 26.7% WER on LRS3 - Finetune (LoRA + encoder): 25.4% WER on LRS3 — only 1.3 point gain **in-domain** - On out-of-domain data, the encoder adaptation gain would be **much larger** because the encoder has never seen the target visual conditions

Expected improvement: 15-25 WER points (from ~67% to ~42-52%)

3.4 Summary: Expected Impact

Configuration	Trainable Params	Expected WER	Improvement
Current (no fine-tuning)	0	67.0%	Baseline
LoRA only (freeze)	~4.2M	~59-64%	3-8 points
LoRA + Bridge (freeze)	~8.4M	~57-62%	5-10 points
LoRA + Bridge + Encoder (FT)	**~323M**	**~42-52%**	**15-25 points**

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4. Is This the Same Training Method as the Paper?

4.1 Configuration Comparison

Parameter	Paper (Section 4.2)	This Codebase	Match?
Visual encoder	AV-HuBERT Large	AV-HuBERT Large (`large_vox_iter5.pt`)	YES
LLM backbone	LLaMA-2-7B	LLaMA-2-7B	YES
Quantization	QLoRA (4-bit NF4)	QLoRA (4-bit NF4, double quant)	YES
LoRA rank	16	16	YES
LoRA alpha	32	32	YES
LoRA targets	Q, V projections	Q, K, V projections	**Slightly different**
LoRA dropout	5%	5%	YES
Learning rate	5e-4	5e-4	YES
Optimizer	Adam (beta1=0.9, beta2=0.98)	Adam (beta1=0.9, beta2=0.98)	YES
LR schedule	Tri-stage	Tri-stage (10k warmup, 0 hold, 20k decay)	YES
Total steps	30k (for 433h data)	30k	YES
Encoder freeze	18k steps (FT variant)	18k (finetune.yaml)	YES
GPUs	8x RTX 3090 (24 GB each)	1x Tesla T4 (16 GB)	**NO**
Effective batch	1 x 8 GPUs = 8	1 x update_freq 8 = 8	YES (via gradient accumulation)
K-means clusters	200	200	YES
Label smoothing	Not explicitly stated	0.1	Likely match
Training data	LRS3 (433h)	AVSpeech (YouTube)	**Intentionally different**

4.2 The LoRA Target Difference

The paper targets Q and V projections. The codebase also targets K. This adds ~30% more LoRA parameters but is generally harmless or slightly beneficial. The codebase implementation is a reasonable enhancement.

4.3 Recommended Modifications for Domain Adaptation

For fine-tuning on AVSpeech (domain adaptation), three changes to the paper's method are recommended:

Change 1: Start from existing checkpoint (not from scratch) - The paper trains from pre-trained AV-HuBERT + fresh LoRA - For domain adaptation, start from `checkpoint_finetune.pt` (already LRS3-trained) - This is standard transfer learning — preserves lip-reading ability, adds new domain

Change 2: Reduce encoder freeze duration - Paper: freeze for 18k of 30k steps - Recommended: freeze for **0-5k steps** only - Rationale: The encoder IS the bottleneck for domain adaptation — we want it to adapt as early as possible - The short freeze gives LoRA/bridge time to warm up before encoder gradients propagate

Change 3: Scale training steps to data size - Paper: 30k steps for 433h of data - If AVSpeech subset < 100h: **10k-15k steps** to avoid overfitting - Scaling formula: `steps = 30k * (data_hours / 433h)` - Risk: Overfitting on small diverse dataset is worse than underfitting

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5. GPU Requirements: Tesla T4 vs. 8x Large GPU

5.1 Current Server Hardware

Component	Specification
GPU	Tesla T4 (15,360 MiB / 16 GB VRAM)
GPU Compute	8.1 TFLOPS FP32, 65 TFLOPS FP16 (Tensor)
CPU	Intel Xeon Platinum 8259CL @ 2.50GHz
RAM	30 GB
Disk	157 GB free

5.2 Training Memory Budget (T4, 16 GB)

Component	VRAM Usage (estimated)
LLaMA-2-7B (4-bit quantized)	~3.5 GB
AV-HuBERT encoder (FP16)	~0.6 GB
Bridge layer + LoRA adapters	~0.05 GB
Subtotal: model weights	**~4.1 GB**
Forward activations (batch=1, 500 frames)	~2-4 GB
Backward gradients (encoder unfrozen)	~3-5 GB
Optimizer states (Adam momentums)	~1-2 GB
CUDA overhead	~1-2 GB
Total estimated	**~11-17 GB**

5.3 T4 Feasibility

Phase	Fits in 16 GB?	Notes
Encoder frozen	YES (comfortable)	~8-10 GB total
Encoder unfrozen	**TIGHT** (may OOM)	~14-17 GB, may need max_sample_size reduction
Estimated training time	~25-50 hours	Very slow due to single GPU + gradient accumulation

5.4 8x Large GPU Instance (RECOMMENDED)

Instance	GPUs	VRAM/GPU	Total VRAM	Cost/hr	Training Time
p3.16xlarge	8x V100	16 GB	128 GB	~$24/hr	~3-5 hours
p4d.24xlarge	8x A100	40 GB	320 GB	~$32/hr	~2-3 hours
g5.48xlarge	8x A10G	24 GB	192 GB	~$16/hr	~3-4 hours
p5.48xlarge	8x H100	80 GB	640 GB	~$98/hr	~1-2 hours

5.5 Why 8x GPU Is Dramatically Better

1. **Matches the paper's exact setup** — the config already has `distributed_world_size: 8`. No code changes needed.
2. **No memory constraints** — 24+ GB per GPU means unfrozen encoder fits comfortably

3. **Training time: ~3-5 hours vs. ~25-50 hours** — enables iterating on multiple experiments per day
4. **Cost: ~$120-200 per run** on p3.16xlarge — cheaper than 2 days of continuous T4

5.6 Recommended Workflow

Task	Where to Run	Why
Data preprocessing (stages 1-7)	**T4 instance**	CPU-bound, T4 handles it fine
Model training (stage 8)	**8x GPU instance**	Multi-GPU for speed, no memory limits
Evaluation / decode	**T4 instance**	Single-GPU decode works fine
K-means re-clustering	**Either**	GPU helps but not critical

The config only needs one change for 8x GPU:

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optimization:
  update_freq: [1]   # Change from [8] — no accumulation needed with 8 real GPUs
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6. The AVSpeech Data Challenge

6.1 AVSpeech Dataset Overview

AVSpeech (Ephrat et al., Google, SIGGRAPH 2018) is a large audio-visual dataset: - ~4,700 hours from ~290,000 YouTube videos - Segments of 3-10 seconds, single visible speaker - Wide variety of people, languages, face poses - **No transcriptions provided** — designed for speech separation, not recognition

6.2 Generating Training Labels

Since VSP-LLM trains with `(video, transcription)` pairs, AVSpeech needs transcriptions. The best approach:

1. **Run Whisper ASR on AVSpeech audio** to generate transcriptions
2. This is already built into the pipeline (Stage 4: `lib/asr.sh`)
3. Quality: Whisper achieves ~5-10% WER on clean speech, higher on noisy
4. **This creates a noise ceiling** — the lip-reading model cannot exceed label quality

6.3 Is Noisy Labeling Acceptable?

Yes, for domain adaptation. The key insight:

> Noisy labels from many domains > perfect labels from one domain

The current model has perfect LRS3 labels but catastrophically fails on YouTube because it never saw diverse visual conditions. Even with 10% label noise from Whisper, the visual encoder would learn to handle: - Side profiles and moving heads - Poor lighting and webcam quality - Diverse lip shapes and speaking styles

This is validated by LiteVSR (ICASSP 2024), which achieved competitive results using only Whisper-generated labels for training.

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7. What Fine-Tuning Will and Won't Fix

7.1 Failure Modes Addressed

Failure Mode	Current Impact	Helped by Fine-Tuning?	Explanation
Visual domain mismatch	~all segments	**YES (strongly)**	Encoder learns diverse visual conditions
Vocabulary mismatch	~60% of segments	**YES (moderately)**	LoRA learns YouTube vocabulary
Named entity errors (F1: 38.8%)	85% of segments	**Partially**	More diverse vocabulary in training data, but proper nouns remain hard
Insertion errors (Corpus WER 125.5%)	Widespread	**Partially**	Better visual features reduce ambiguity, fewer hallucinated insertions

7.2 Failure Modes NOT Addressed

Failure Mode	Current Impact	Helped by Fine-Tuning?	Why Not
Hallucination (WER >= 100%)	20.6% of segments	**No**	Architectural — the LLM generates from language prior when visual features are ambiguous, regardless of training data
Short segment failure (1-5 words)	~8% of segments	**No**	Architectural — the LLM needs context length; more training data doesn't add context at inference time

Failure Mode	Current Impact	Helped by Fine-Tuning?	Why Not
Homophene ambiguity	Fundamental	**Minimally**	Many words look identical on the lips (/p/, /b/, /m/). No amount of training changes lip physics.

7.3 Realistic Outcome

Metric	Current	After Fine-Tuning (est.)	Improvement
Mean WER	67.0%	~42-52%	15-25 points
Corpus WER	125.5%	~85-105%	20-40 points
Perfect (WER=0)	1.9%	~4-8%	2-4x more
Good (WER <= 20%)	11.4%	~18-28%	2-2.5x more
Catastrophic (WER >= 100%)	20.6%	~12-18%	Moderate reduction
NEA F1	38.8%	~45-55%	6-16 points

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8. Alternative and Complementary Approaches

8.1 Approaches That Complement Fine-Tuning

These can be combined with fine-tuning for cumulative improvement:

Approach	Est. WER Reduction	Effort	Combined with FT?
Hyperparameter tuning (Report 2)	10-15 points	Low	Yes (independent)
Prompt engineering (Report 3)	5-10 points	Low	Yes (independent)
Confidence scoring (Report 4)	Enables filtering	Medium	Yes (post-training)
N-best aggregation (Report 5)	5-10 points	Medium	Yes (post-training)
All above + fine-tuning	**25-40 points**	High	Cumulative

8.2 Alternative Training Approaches (Literature)

Method	Source	Data Required	Expected Impact
Whisper knowledge distillation	Prajwal et al., Interspeech 2024	Unlabeled AV data	5-10 WER points (complementary to QLoRA)
Speaker-adaptive prompt tuning	Shin et al., AAAI 2025	5 min per speaker	Significant per-speaker gains
MultiVSR dataset fine-tuning	Prajwal et al., ICASSP 2025	12k hrs labeled	Potentially stronger than AVSpeech (pre-labeled)
VALLR phoneme decomposition	Thomas et al., ICCV 2025	Architectural change	Addresses hallucination at architecture level

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9. Implementation Roadmap

Phase 1: Data Preparation (This T4 instance, ~1-2 days)

1. Transfer AVSpeech videos to this instance (`/home/ubuntu/vsp_input/`)
2. Run existing pipeline stages 1-7 (normalize, segment, crop, ASR, manifests, clustering)
3. Verify training data format (train.tsv, train.wrd, train.km, train.cluster_counts)

Phase 2: Training (8x GPU instance, ~3-5 hours)

1. Transfer processed data to 8x GPU instance
2. Run fine-tuning with AVSpeech config:
3. Start from `checkpoint_finetune.pt`
4. Freeze encoder for 5k steps, unfreeze for 10k steps
5. `update_freq: [1]` with 8 real GPUs
6. Save best checkpoint based on accuracy metric

Phase 3: Evaluation (This T4 instance, ~2-4 hours)

1. Transfer fine-tuned checkpoint back
2. Re-run decode on english_1k with new checkpoint
3. Generate report, compare WER/NEA against baseline
4. If improvement is sufficient, deploy; otherwise iterate

Phase 4: Post-Training Optimization (optional, ~1 day)

1. Re-run K-means clustering with fine-tuned encoder features

2. Regenerate cluster counts
3. Re-decode with new clusters
4. Apply hyperparameter tuning from Report 2

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10. Conclusion

Fine-tuning on AVSpeech data is **the right strategic move** for improving real-world performance. The expected 15-25 WER point improvement (from ~67% to ~42-52%) would make the model meaningfully more useful, though still far from perfect.

Key decisions: - **Unfreeze the encoder** — this is where 70% of the gain comes from - **Use 8x GPU instance** for training — 10-20x faster, no memory constraints, matches paper's setup - **Start from existing checkpoint** — transfer learning preserves LRS3 knowledge - **Generate labels with Whisper** — noisy but acceptable for domain adaptation - **Combine with hyperparameter tuning and confidence scoring** for maximum cumulative improvement

The model's fundamental limitation — LLM-based decoding causes hallucination when visual features are ambiguous — cannot be solved by training data alone. For production reliability, confidence scoring (Report 4) and N-best aggregation (Report 5) remain essential regardless of fine-tuning.

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11. References

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